SAGAR KUMAR SAHU

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SUMMARY

Backend-focused Software Engineer skilled in Java, Spring Boot, and Microservices, with hands-on experience building secure REST APIs, JWT-based authentication, and scalable backend systems. Comfortable designing services with caching, concurrency control, and sound system design principles. Strong foundation in data structures, algorithms, and system design, backed by a GATE 2026 (CS) qualification and a consistent academic record (CGPA: 9.07/10). A quick learner who enjoys picking up new backend tools and frameworks, and actively contributes to open-source projects outside of coursework.

EDUCATION

VIT Bhopal University *2023 – 2027 (Expected)*

B.Tech in Computer Science and Engineering – CGPA: 9.07/10

Relevant Coursework: Data Structures & Algorithms, OOP, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, System Design

TECHNICAL SKILLS

Languages: Java, Python, SQL

Frontend: HTML5, CSS3

Backend: Spring Boot, Spring MVC, Spring Security, Spring AI, JWT (JJWT), Hibernate, JPA, REST APIs, GraphQL

Systems & DevOps: Microservices, System Design, Distributed Systems, Spring Cloud, Kafka, Redis, Docker, Optimistic Locking, Git & Version Control, GitHub Actions

Databases: PostgreSQL, MySQL, MongoDB

Tools: Maven, Postman, Lombok

EXPERIENCE

Bluestock Fintech *May 2025 – Jun 2025*

Software Development Engineer (SDE) Intern

- Contributed to backend development of platform features, working on REST API modules in an agile engineering team.
- Collaborated on debugging, testing, and performance improvements for production backend application modules.
- Participated in code reviews and daily stand-ups, gaining exposure to real-world software development workflows in a fast-moving fintech startup.

PROJECTS

Simple RAG Document Q&A System

Java, Spring Boot, Spring AI, Gemini, MariaDB | *GitHub*

- Built a RAG application with Spring Boot and Spring AI where users upload documents and ask questions answered only from the uploaded content.
- Implemented document parsing, chunking, and embedding pipeline, storing vectors in MariaDB Vector Store and retrieving relevant chunks via similarity search.
- Integrated Gemini LLM through Spring AI ChatClient to generate context-grounded answers with source references.

eKart Pro – Multi-Vendor E-Commerce Backend

Java 17/22, Spring Boot 3, Spring Security 6, JWT, PostgreSQL, Hibernate | *GitHub*

- Developed the backend for a multi-vendor e-commerce platform with stateless JWT authentication and RBAC for Customer, Seller, and Admin roles.
- Designed a relational schema of 11 entities (users, products, orders, payments, reviews, carts, wishlists) with foreign-key integrity.
- Implemented transactional order processing (@Transactional) handling stock validation, inventory deduction, payment record creation, and cart clearing.
- Built seller and admin analytics services aggregating revenue, GMV, and order-pipeline status transitions.

AgriRent – Smart Agriculture Equipment Rental Platform

Microservices Architecture, Spring Cloud, Kafka, Redis, MySQL, REST APIs | *GitHub*

- Architected a microservices backend connecting farmers with equipment vendors via Auth, Zone, Vendor, Catalog, Order, Payment, Advisory, Community, and AI Assistant services.
- Designed JWT-based authentication and RBAC (Admin, Vendor, Farmer, Expert) enforced at the API Gateway and service level using Spring Cloud Gateway.
- Planned asynchronous, event-driven communication via Kafka (order.created, order.completed, payment.settled) with a database-per-service pattern.
- Designed vendor-matching logic across Zone and Vendor services with Redis caching for availability lookups.

CERTIFICATIONS & ACHIEVEMENTS

- GATE 2026 (Computer Science) – AIR 10770, Score 461
- Marketing Analytics – NPTEL, IIT Kharagpur (Elite, Top 5%), 2026
- Introduction to Machine Learning – NPTEL, IIT Madras, 2025
- The Bits and Bytes of Computer Networking – Google, via Coursera, 2024
- Solved 250+ DSA problems on LeetCode

ADDITIONAL INFORMATION

Languages: English, Hindi, Odia. Actively follows developments in backend engineering and system design, and enjoys competitive programming and contributing to open-source communities in spare time.